

Measures of Center

Wednesday, February 5, 2025 5:24 PM

In summary:

$$\text{mean } \bar{x} = 68.9$$

$$\text{median } \tilde{x} = 73.0$$

$$\text{mode } M = 73$$

which is Better?

Given a Quantitative (numeric) set of Data,
one can ask the question: What is a typical value of the data?

For Example:

Suppose these are exam scores

$\{7, 64, 67, 70, 72, 73, 73, 73, 76, 82, 85, 85\}$

This question can be answered as follows:

I. The Mean - the average in which you add all the values and divide by the number of values.

$$\mu = 68.9 \quad \bar{x} = 68.9$$

population μ
("mean")

sample $\bar{x}$
("x-bar")

II. The Median - this is the center value in a set of sorted numbers. If an odd number of data, take the exact middle; if an even number of data, take midpoint of the two middle values.

$$\tilde{x} = 73.0$$

"x-kidde"

* NOTICE
ROUNDING

III. The Mode - this is the most frequently occurring value; sometimes there might be two modes and this is called bi-modal. If more than two modes, it is then called multi-modal; however, that has no practical value.

$$M = 73$$

no rounding

Note: throughout this textbook, measures of center are referred to as ~~Typical Values~~; however, that is NOT the terminology used in Statistics. These values are called Measures of Center.

Measures of Center tell us approximately where the center of the data is.

I. The Mean

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scratch

$$59/4 = 14.75$$

FINDING THE MEAN

A **typical value** is an ^{datum} amount that represents the values in a set of numbers. A typical value must fall within the series of numbers it represents. There are three common typical values in a set of numbers: **mean**, **median**, and **mode**.
1. 2. 3.

When most people use the word **average**, they are talking about the **mean**. For a set of values that are similar to each other, the mean is what you want to know when you ask, "About how much . . . ?"

To find the mean, add the values in a set of data, and then divide the sum by the number of values in the set.

FINDING THE MEAN

Example Ella scored 14, 18, 11, and 16 points in her last 4 basketball games. What is her scoring average?

Step 1

Find the mean. Add the values in the set.

$$14 + 18 + 11 + 16 = 59$$

Ella's scoring average is **14.75 points**.

Step 2

Divide the sum by the number of values in the set.

$$\frac{59}{4} = 14\frac{3}{4} = 14.75$$

The mean is not necessarily equal to any number in the set. However, it must fall within the range of the numbers in the set.

$$\{11, 14, 16, 18\} \quad \mu = 14.8$$

Population, Not a Random Sample

Examples of the Mean

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When calculating the mean (average), always write one more decimal place for the mean. For example, if the data are whole numbers, write the mean with one decimal point. The only time you would break this rule is if the extra decimal makes no sense; e.g., a mean of \$1.583 makes no sense! Also, when rounding, if 5 or more, round up; if less than 5, round down.

ASSUME THESE ARE ALL SAMPLES.

Find the mean for each set of numbers.

1. 35, 49, 31, 25

$$\bar{x} = 35.0$$

\$1.25, \$2.08, \$0.97, \$3.12, \$1.48

$$\bar{x} = \$1.78$$

common sense, rounding *

5, 4, 12

$$\bar{x} = 7.0$$

2. 174, 12, 398, 214, 45

$$\bar{x} = 168.6$$

6, 3, 15, 9, 4

$$\bar{x} = 7.4$$

\$35, \$12, \$97, \$6, \$51, \$90

$$\bar{x} = 48.50$$
 *

3. \$215, \$154, \$192

$$\bar{x} = \$181$$
 *

87, 96, 95, 100, 75, 82, 88

$$\bar{x} = 89.0$$

32.1, 15.75, 17.2, 18.35, 28.4

$$\bar{x} = 22.36$$

4. 5.6, 5.1, 6.4

$$\bar{x} = 5.70$$

\$240, \$198, \$264, \$249, \$220

$$\bar{x} = \$234.20$$
 *

1, 7, 13, 7, 10

$$\bar{x} = 7.6$$

Solve.

5. In July, DeShaun's weekly grocery bills were \$84, \$98, \$72, and \$60. What is his mean weekly grocery bill for the month?

NOT A RANDOM SAMPLE

$$\mu = \$78.50$$

Pop: all weeks in a particular month

6. Miranda received scores of 83, 78, 85, 82, and 77 on 5 tests. What is her average score on the 5 tests?

Not sure if these are all her tests μ or just a sample $\bar{x}$

$$81.0$$

Examples of the Mean, cont'd...

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If you know the mean, you can use algebra to find a missing value. You can also use the mean to draw conclusions and make predictions.

There are called:

CONFIDENCE INTERVALS
& HYPOTHESIS TESTING

USING THE MEAN

Example 1 Sean budgets \$400 a month for food. In 5 months, he spent \$415, \$380, \$489, \$392, and \$404. In order to average \$400 per month, how much should he spend in the sixth month for food?

Step 1

Write an equation to represent the mean. Let x represent the amount spent in the sixth month.

$$\frac{415 + 380 + 489 + 392 + 404 + x}{6} = 400$$

Sean should spend **\$320** in the sixth month to average \$400 per month.

Example 2 Gina is paid weekly based on the amount of sales she makes. Her first paychecks are for the following amounts: \$346, \$358, \$320, and \$336. Based on the mean for the first 4 weeks, about how much will Gina make in a year?

Step 1

Find the mean of the first 4 weeks.

$$\frac{346 + 358 + 320 + 336}{4} = 340$$

$$\mu = \$340$$

Gina will make about **\$17,680** in a year.

Step 2

Solve for x .

$$\frac{415 + 380 + 489 + 392 + 404 + x}{6} = 400$$

$$\frac{2,080 + x}{6} = 400$$

$$2,080 + x = 2,400$$

$$x = 320$$

special note

if there are n data

then there are $n-1$

Degrees of Freedom

$\therefore n=6$ means
 $df=5$ } there is only
one possible
answer for x .

Step 2

Gina will typically make \$340 a week. There are 52 weeks in a year. Multiply the mean by 52.

$$\$340 \times 52 = \$17,680$$

yearly income $\approx \$17,680$

Examples of the Mean, cont'd...2...

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Solve as directed.

7. Jena plans to run an average of 6 miles a day. For the first 6 days, she runs 3, 8, 8, 0, 4, and 5 miles. How many miles should she run on the seventh day to meet her goal?

$$\mu = 6.0$$

$$\frac{3+8+8+0+4+5+x}{7} = 6$$

$$x = 14 \text{ miles}$$

8. In the first 8 weeks of school, students at Pine Elementary read the following numbers of books.

80 85 92 78 80 81 90 96

Based on the average of the first 8 weeks, how many books will they read in a 36-week school year?

$$\mu = 85.25$$

$$= 85.3 \text{ per wk}$$

$$36 \text{ wks} (85.3) = 3070.8$$
$$= 3071$$

9. Jared set a goal to sell 300 alarm systems in a 60-day period. At the end of the 60 days, he had averaged 4 alarms per day. Did he meet his goal?

$$60 \times 4 = 240$$

NO.

10. **Explain** Carmel received test scores of 77, 81, 95, and 91. She has 1 more test to take. In order to get an A in the class, she needs an average of 90 on the 5 tests. If the highest score Carmel can earn on the last test is 100, is it possible for her to get an A in the class? Explain your thinking.

$$\frac{77+81+95+91+x}{5} = 90$$

$$x = 450 - 344$$

$$x = 106$$

Not possible!!